

The Soviet scheme was to simultaneously launch these missiles from submarines and bombers against the carrier from very long distances. Navigation computers of these missiles were programmed in such a way that all cruise missiles reached the carrier at the same time but from different directions. This was called the saturation cruise missile strike plan of the Soviet. Even if one cruise missile would hit the target, that would have been sufficient. This changed the situation in favour of the USSR.

Air defence warfare capabilities of the CSG have been on constant updates and upgrades all these years. Today, air defence systems of an American CSG can be termed as the most superior system in the world. Protecting the CSG from cruise missiles and cruise-missile-launching aircraft is called air defence warfare.

Air defence operations are carried out in four phases.

### Phase I: Fighter aircraft and AEW aircraft combo

The first line of defence of the CSG is carrier-borne F-18 combat aircraft, which can operate far away from the carrier and shoot down enemy bombers before they launch cruise missiles. But these cannot find bombers quickly, because they do not know their exact location.

Onboard radars of F-18 aircraft can accurately scan specific sectors and specific altitudes but not all sectors and altitudes. To guide these towards incoming bombers, a dedicated airborne early warning (AEW) aircraft (also carrier-borne) is used.

These AEW aircraft, named E2, have huge radars. An inverted saucer-like structure on the top is the radome for the radar antenna. Radome is a structure made of radio-transparent material to protect the radar antenna from the environment. Inside this radome is the rotating antenna of the high-power UHF Doppler radar.

This radar can detect and track multiple threats on the sea and in



Fig. 3: AN/SPS-49 radar antenna

the air. It can simultaneously and automatically detect and track 2000 targets, spread across the air and sea.

The aircraft keeps tab on an area of over twelve million cubic kilometres (three million cubic miles). Not only that, it can precisely track cruise missiles that fly low enough, as if skimming the water surface. Being a crucial component for the defence of the carrier, this radar is highly capable of resisting the jamming efforts of the enemy.

These aircraft maintain a surveillance patrol far away from the carrier and act as a command-and-control (C<sup>2</sup>) station. These see what the radars of the CSG cannot see. If AEW aircraft detect any enemy bombers at a long range, these communicate the same to the carrier through a data link. Immediately, F-18 combat aircraft from the carrier are scrambled. These take off and race towards incoming enemy bombers to intercept and stop these from launching deadly cruise missiles. When F-18 aircraft reach their area of responsibility, AEW aircraft vector these towards the threat. F-18s intercept and shoot down bombers. There are four such AEW aircraft on a carrier.

### Phase II: Air search radars and fighter aircraft combo

Air defence operations relying on this combination is a grave scenario. It occurs when the AEW aircraft itself has been shot down. This scenario is very much possible if the incoming enemy aircraft force is large. In

this case, CSG radars are the only means to detect enemy air targets. For this, the carrier has two air surveillance radars: a 2D radar and a 3D radar.

#### 2D air-search radar.

A 2D radar gives the two dimensions of the target's position: range and bearing of the target. 2D radar AN/SPS-49 operates in L band (851MHz–942MHz) and can detect

air targets at a range of 463km (250 nautical miles).

It can scan up to an altitude of 46km. A two-cavity Klystron oscillator is behind its peak power of around 350kW. Due to the thinness of the beam produced, it becomes difficult for the enemy to jam it.

Its antenna rotates at 6rpm to detect flying objects at longer ranges. The flying object's position is updated every ten seconds. When that flying object approaches further, the antenna rotates at 12rpm. Now, the flying object's position gets updated every five seconds, which results in the target being tracked continuously.

This radar can simultaneously track 255 targets. Since it is sitting on the ship, it is subjected to the pitch and roll of the ship. To overcome this, the antenna is gyro-stabilised. And because of this, even if the ship rolls to the side or pitches up or down due to the waves, this antenna stays where it was before the roll or pitch.

Through identify friend or foe (IFF) components organic to the radar, the targets' friendliness is determined.

IFF shoots an ultra-short pulsed narrow beam of radar energy to the flying objects as a challenge. The friendly aircraft through an IFF receiver receives this and responds with a suitable reply. The radar receives this reply and classifies the flying objects as friendly aircraft. Now for the radar's IFF challenge, if there is no proper reply from an air-